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European Patent Application No 22807450.6
In the name of Bridgestone Corporation
ESTIMATION DEVICE, ESTIMATION METHOD, PROGRAM, AND TRAINED MODEL GENERATION DEVICE

In response to the communication pursuant to Rules 70(2) and 70a(2) EPC dated 18 October 2024, we hereby confirm that the applicant wishes to proceed further with the application.

Also in response to the above-mentioned communication, we enclose the following:

1. An amended set of claims, comprising claims 1 to 8, to replace the set currently on file; and
2. A marked-up version of the claims, to indicate clearly the amendments made.

Novelty and Inventive Step

At Item 2 of the extended European search report the examiner argued that the subject matter of claim 1 lacks novelty. At Item 3.1 the examiner argued that the subject matter of claim 2 does not involve an inventive step. However, the examiner then stated that:

"The application appears to provide basis (fig. 5, 7, 15, [0036], [0038], [0085]) for amending the claim such that indirect measurement of the physical state of the system is performed by performing measurement of the input quantities (pressure and resistance) through technical means (various sensors as per [0036]). In such a hypothetical claim the differences identified above will make technical contribution to the prior art."

Claim 1 has now been amended to add the features of claim 2. In addition, claim 1 has been amended according to the examiner's suggestion at Item 3.1 to add the following additional wording:

> "and wherein the learning model is trained using:
> a pressure sensor to detect pressure as the first physical quantity;
> an electrical characteristic detection section (76) that includes a sensor to detect electrical resistance values as the second physical quantity."

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This amendment has basis in paragraph [0036]. We submit that it is not necessary to add the other features discloses in the embodiment described by paragraph, for the following reasons.

Firstly, although paragraph [0036] describes the features of the embodiment in the context of a rubber actuator 2, it is made clear in the description at, for example, paragraphs [0025] and [0120] that the rubber actuator 2 is merely an example of an application target. Therefore, it is not necessary to add the rubber actuator 2 to claim 1. In addition, the various features of the rubber actuator 2 (such as the closure members 23) have not been added to claim 1 either.

Next, the distance sensor 77 has not been added to claim 1 because paragraph [0037] makes clear that it is not necessary to identify the length of the rubber actuator 2.

In addition, the counter 80 has not been specifically defined in claim 1 because paragraph [0038] makes clear that the counter 80 is not essential. Instead, the wording of original claim 2 remains which defines that the third physical quantity indicates a number of repetitions of deformation of the member.

Furthermore, we submit that the validation described in paragraph [0085] with respect to Fig. 15 is also not essential to the measurement of pressure and electrical resistance features added to claim 1, and thus the features of paragraph [0085] and Fig. 15 have not been added to claim 1.

As noted by the examiner in the section reproduced above, a claim in which indirect measurement of the physical state of the system is performed by performing measurement of the input quantities (pressure and resistance) through technical means, means that the differences identified above will make technical contribution to the prior art. Accordingly, we submit that the differences between the subject matter of amended claim 1 and the prior art make a technical contribution to the prior art. We therefore submit that the subject matter of amended claim 1 is novel, technical and non-obvious from the cited documents. Therefore, we submit that amended claim 1 is allowable.

Independent method claim 8 and program claim 9 (new claims 7 and 8) have been amended in a corresponding way to claim 1, and are allowable for corresponding reasons.

Article 84 EPC

Claim 10 has been deleted to address the objection at Item 4.1.

Other Matters

The claims have been provided with reference signs, as requested.

Claims 1, 7 and 8 have not been written in the two-part form because it is considered that doing so would reduce the clarity of the claims, contrary to Article 84 EPC.

We request amendment of the description be deferred until the current claims have been examined.

Conclusion

In view of the above, we believe that the claims are allowable. Favourable reconsideration is therefore requested.

We take the precaution of requesting oral proceedings by video-conference before any decision to refuse the application is issued.

We reserve the right to reintroduce any subject matter deleted from this application by way of amendment or by way of filing one or more divisional applications.

/Signed electronically/

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